Supervised and Self-Supervised Semantic Segmentation for Aircraft Emergency Landing-Site Identification

Fahmida Jannat, Sumaiya Akter, and Nusrat Ahmed Sruti
Department of Computer Science and Engineering, East West University
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Instructor: Dr. Md Rifat Ahmmad Rashid

***Abstract*—Identifying a safe emergency aircraft landing site from aerial imagery is a difficult visual task because suitable terrain must be identified at the pixel level rather than simply detected as an object. This project evaluates fully supervised and self-supervised semantic-segmentation approaches on the Moroccan Aircraft Emergency Landing Sites dataset. In Part A, DeepLabV3–ResNet50, SegFormer-B0, and YOLO26n-Sem were trained using the same leakage-safe split; DeepLabV3–ResNet50 achieved the best supervised mIoU of 0.8220. In Part B, SimCLR, BYOL, MAE, and DINOv2 learned visual representations before segmentation fine-tuning with only 314 labelled images, compared with 1,416 labelled images for the supervised baseline. MAE achieved the strongest SSL result, reaching 0.8055 mIoU and recovering 98.00% of the baseline performance with 77.82% fewer labelled images.**

***Index Terms*—Semantic segmentation, aerial imagery, self-supervised learning, DeepLabV3, SimCLR, BYOL, MAE, DINOv2.**

I. Introduction

Semantic segmentation assigns a class label to every image pixel. This is important for aerial emergency landing-site identification because a usable region must be localized with an accurate extent and boundary, rather than only detected at image level. The present task is binary semantic segmentation with two classes: background and landing site.

Pixel-level annotation is expensive and time-consuming because a human must mark the correct region boundary for every training image. This creates a practical limitation for remote-sensing applications, where unlabelled aerial images may be easy to collect but accurate masks are costly to obtain. The project was therefore divided into two complementary parts.

Part A establishes a strong fully supervised benchmark using DeepLabV3–ResNet50, SegFormer-B0, and YOLO26n-Sem. These models represent CNN-based, transformer-based, and lightweight segmentation approaches under the same experimental conditions. Part B investigates label efficiency: SimCLR, BYOL, MAE, and DINOv2 are first pretrained from images without segmentation labels, then adapted using a small labelled subset. The central question is how much supervised segmentation performance can be retained when the number of manually labelled images is substantially reduced.

This comparison is relevant to emergency-planning applications because a false positive could mark unsuitable terrain as safe, while a false negative could exclude a viable landing area. In practice, the model should therefore be judged not only by the number of correctly classified pixels, but also by the continuity and location of its predicted landing-site boundaries. The report consequently combines numerical metrics with qualitative failure analysis.

II. Dataset and Preparation

The Moroccan Aircraft Emergency Landing Sites dataset contains 2,090 valid image–mask pairs, each at 256×256 pixels. RGB masks were verified and converted to single-channel class-index masks: 0 for background and 1 for landing site. File matching, corrupted-file checks, dimension checks, and duplicate analysis were performed before model training.

The dataset has 1,570 landing-site-positive images and 520 background-only images. Background accounts for 69.81% of pixels and landing sites account for 30.19%, a moderate 2.31:1 imbalance. For this reason, mIoU and Dice score are emphasized alongside pixel accuracy.

Each ground-truth mask is treated as a dense supervision signal: every pixel contributes either to the background class or the landing-site class. Converting the original colour masks to class indices simplifies this representation and ensures that all model outputs are evaluated under the same binary convention. The background-only images are also important because they discourage a model from predicting a landing region in every aerial scene.

TABLE I
Data-quality validation summary.

Validation check	Problem count
Corrupted images or masks	0
Missing image–mask pairs	0
Dimension mismatches	0
Unknown mask colours	0
Images/masks not 256×256	0

Table I confirms that the usable dataset is clean and internally consistent. Because no files were corrupted, unmatched, or incorrectly encoded, later performance differences are unlikely to be caused by data-quality errors.

TABLE II
DATASET COMPOSITION.

Image type	Count	Share
Landing-site positive	1,570	75.12%
Background only	520	24.88%
Total	2,090	100.00%

Table II shows that most images contain at least one landing-site region, while the background-only examples ensure that models also learn when no safe region should be predicted.

TABLE III
FIXED DATASET SPLIT.

Partition	Images	Role in Part B
Training	1,416	SSL pretraining without masks
Validation	314	Labelled downstream fine-tuning
Test	360	Monitoring and final evaluation

Table III shows that all methods use the same fixed partitions. In Part B, the training images remain unlabelled during SSL pretraining, while only the 314-image validation partition provides labels for downstream adaptation.

A seed of 42 was used and the exact split was reused by every model. Training images were resized and normalized. Flips, rotations, brightness/contrast adjustment, and Gaussian blur were applied only during training; geometric transforms were applied jointly to images and masks to preserve alignment.

The fixed split is particularly important because small changes in the number or difficulty of labelled samples can strongly influence measured SSL performance. Reusing identical partitions makes the comparison attributable to the representation-learning method rather than to different examples seen during training. Augmentation also exposes models to moderate changes in viewpoint and illumination without changing the intended class of a pixel.

No augmentation was applied to validation or test images. Keeping evaluation inputs unchanged provides a stable estimate of generalization and avoids artificially improving the final scores through multiple transformed versions of the same test scene. Normalization was applied consistently across the training and evaluation pipelines so that intensity scale did not become a source of variation between models.

A. Evaluation Metrics

For each class, Intersection over Union measures the overlap between prediction and ground truth:

$$IoU = \frac{TP}{TP + FP + FN}. \tag{1}$$

The primary metric, mIoU, is the mean IoU across background and landing-site classes. Dice score is also reported because it emphasizes overlap between the predicted and true landing-site regions:

$$Dice = \frac{2TP}{2TP + FP + FN}. \tag{2}$$

Pixel accuracy is retained as a supporting metric, but is interpreted carefully because background pixels are more common.

For a binary mask, a prediction can have high pixel accuracy simply by classifying many background pixels correctly. mIoU is more informative here because false positives and false negatives both reduce the overlap score. Mean Dice provides a complementary measure of region agreement; together, these metrics make it easier to distinguish a generally accurate model from one that only benefits from the dominant background class.

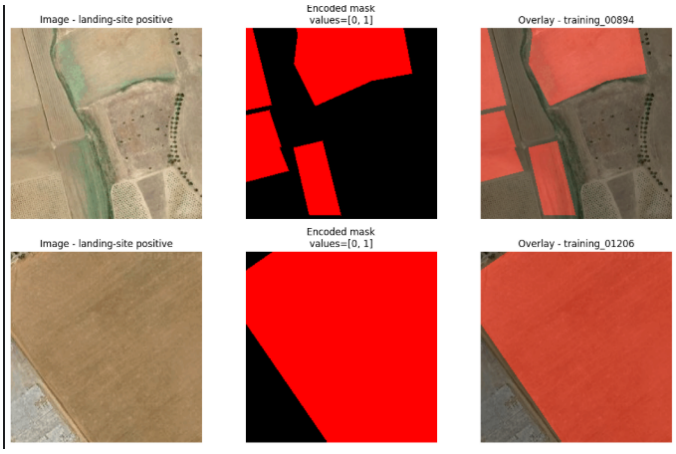


Fig. 1. Representative aerial images, binary masks, and overlays.

Figure 1 illustrates the visual diversity of the aerial scenes and the shape variation of candidate landing-site regions. The overlay panels verify that the converted binary masks align with the corresponding image structures before model training.

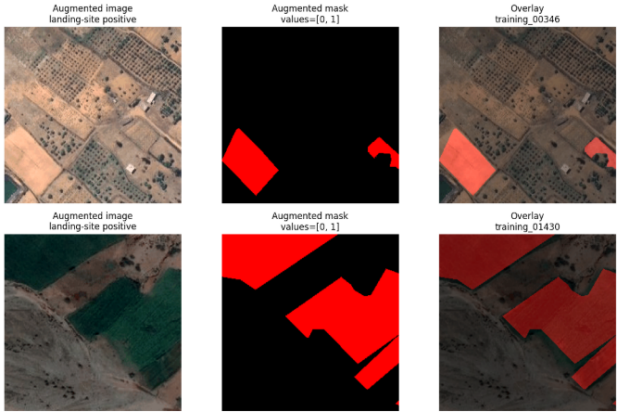


Fig. 2. Examples of training augmentation. Geometric operations preserve image–mask alignment.

Figure 2 confirms that the same spatial operation is applied to an image and its mask, while appearance-only transforma-

tions affect only the aerial image. This prevents the model from learning from misaligned labels.

III. Methods

A. Fully Supervised Models

DeepLabV3–ResNet50 uses atrous convolution and atrous spatial pyramid pooling for multi-scale segmentation. Dilated convolution enlarges the receptive field without excessive loss of spatial resolution, while multi-rate pooling captures landing-site regions at different scales. SegFormer-B0 uses a lightweight transformer encoder–decoder design. Its hierarchical encoder learns multi-scale representations through attention, and its decoder combines them for dense prediction. YOLO26n-Sem offers a lightweight semantic-segmentation alternative based on multi-scale feature extraction and feature fusion. All three models used the same split, enabling a fair comparison.

The supervised models were evaluated using mIoU, class-wise IoU, pixel accuracy, and mean Dice score. DeepLabV3–ResNet50 achieved the best score across all primary metrics. SegFormer-B0 remained close to the best model, suggesting that transformer representations are effective for this aerial segmentation task. YOLO26n-Sem achieved slightly lower accuracy but remained competitive for a lightweight architecture.

The three architectures were selected to create a useful baseline range. DeepLabV3 emphasizes dense multi-scale convolutional features; SegFormer tests a transformer-based alternative; and YOLO26n-Sem represents a compact design where lower computational cost may be valuable in a deployment setting. Since all models receive the same data partition and evaluation procedure, their result differences can be interpreted as architecture and optimization differences rather than split effects.

The supervised benchmark is necessary for interpreting the SSL experiments. Without it, a high SSL score would be difficult to evaluate in context. DeepLabV3–ResNet50 therefore serves as the reference upper bound under the available labelled-data budget, whereas the SSL models test whether much of that performance can be recovered after reducing the labelled fine-tuning set.

B. Self-Supervised Learning

SimCLR learns view-invariant features through a contrastive NT-Xent objective [1]. BYOL uses online and target networks without explicit negative pairs [2]. MAE masks 75% of image patches and reconstructs the missing content [3]. DINOv2 uses student–teacher self-distillation to learn robust visual features [4].

All SSL encoders were pretrained for 50 epochs using the 1,416 training images without masks. Each was then connected to a segmentation decoder and fine-tuned for 50 epochs on the 314 labelled images. Full fine-tuning was selected because SimCLR achieved approximately 0.6966 mIoU with a frozen encoder but 0.8018 mIoU when the encoder was fine-tuned.

The SSL methods differ in how they construct a learning signal from images. SimCLR explicitly compares positive and negative pairs and benefits from strong image augmentations. BYOL avoids negative pairs by predicting the representation generated by a slowly updated target network. MAE learns global visual structure by reconstructing heavily masked images. DINOv2 learns through self-distillation between student and teacher representations. For MAE and DINOv2, transformer patch tokens were rearranged into a two-dimensional feature grid before they entered the segmentation decoder.

Using the same downstream protocol is essential for this comparison. Every SSL encoder sees the same unlabelled images during pretraining and receives the same number of labelled examples during fine-tuning. Thus, the result measures how well each pretraining objective transfers to pixel-level prediction. The frozen-versus-fine-tuned SimCLR result also shows that general visual features need task-specific adaptation to capture precise landing-site boundaries.

Fine-tuning updates the pretrained encoder together with the segmentation decoder. This is useful because pretraining objectives learn broad visual regularities, whereas the downstream task requires a specific decision at every pixel. A representation that recognizes aerial textures can still require adaptation to identify the exact spatial extent of a potential landing region. The common 50-epoch pretraining and 50-epoch fine-tuning schedule further keeps the computational budget comparable among SSL methods.

TABLE IV
SSL objectives used in Part B.

Method	Pretraining objective
SimCLR	Contrastive NT-Xent learning
BYOL	Negative-free online–target prediction
MAE	Masked image reconstruction
DINOv2	Student–teacher self-distillation

Table IV summarizes the distinct learning signals used by the four SSL approaches. This variety makes the comparison useful because the methods learn image structure through contrastive, predictive, reconstructive, and distillation-based objectives.

TABLE V
Shared experimental configuration.

Setting	Value
Input resolution	256×256
Classes	2
Random seed	42
SSL pretraining images	1,416
Fine-tuning labels	314
Test images	360
SSL / fine-tuning epochs	50 / 50
Primary metric	mIoU

Table V lists the conditions shared across the experiments. Keeping the image resolution, data split, epoch count, and evaluation metric consistent supports a fair comparison of the model architectures and pretraining strategies.



Fig. 3. SSL sanity checks before downstream fine-tuning. The views confirm that each pretraining pipeline receives the intended input transformation or reconstruction task.

Figure 3 is a pipeline-level validation step. It confirms that the self-supervised inputs are constructed correctly before segmentation labels are introduced: contrastive methods receive multiple augmented views, whereas masked-image modelling receives partially hidden patches for reconstruction. These checks help rule out preprocessing mistakes that could otherwise make an SSL comparison unreliable.

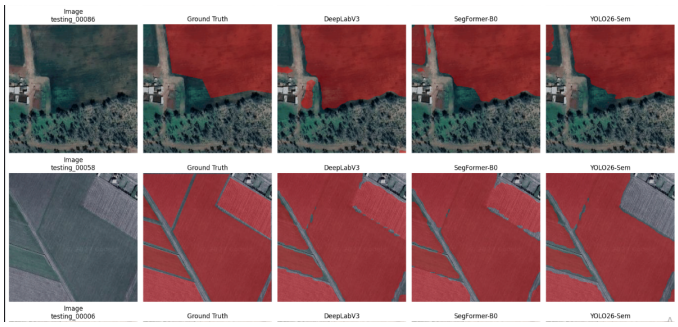


Fig. 4. Representative Part-A predictions. Boundary and small-region differences are visible across supervised models.

Figure 4 compares the three supervised models on identical test samples. All methods locate the principal landing-site region, but DeepLabV3–ResNet50 follows boundaries more consistently, while the other predictions show greater disagreement on narrow or fragmented regions.

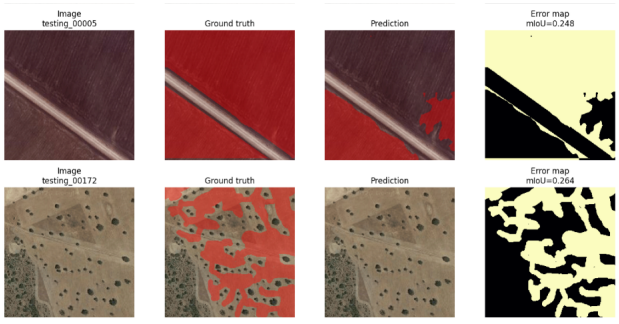


Fig. 5. Part-A low-IoU examples. Errors primarily occur at fragmented or visually ambiguous landing-site boundaries.

Figure 5 shows that the strongest supervised predictions can still fail when target regions are narrow, disconnected, or visually similar to surrounding terrain. These cases motivate the SSL error analysis presented later.

IV. Results and Analysis

Table VI summarizes all final results. DeepLabV3–ResNet50 was the strongest fully supervised method. Among SSL methods, MAE ranked first, followed by SimCLR, DINOv2, and BYOL. Landing-site IoU is lower than background IoU for every SSL method, showing that target boundaries and visually similar terrain are the main source of error.

Table VI shows that DeepLabV3–ResNet50 is the strongest fully supervised model. MAE is the best SSL model on every reported SSL metric, while SimCLR and DINOv2 remain close competitors. BYOL has the largest gap from the supervised baseline but still achieves useful performance with limited labelled data.

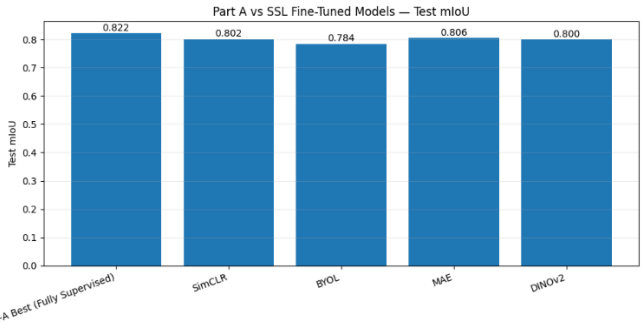


Fig. 6. Test mIoU for the fully supervised baseline and SSL models.

Figure 6 gives a compact ranking of all final models. DeepLabV3–ResNet50 remains the overall best model, while MAE is the best SSL method and performs very close to the supervised baseline despite using substantially fewer labelled images.

MAE has an absolute mIoU gap of $0.8220 - 0.8055 = 0.0165$ from DeepLabV3–ResNet50. It recovers

$$\frac{0.8055}{0.8220} \times 100 = 98.00\%$$

of the baseline mIoU while using only $314/1416 = 22.18\%$ of the labelled training data, a 77.82% reduction. This result shows that MAE provides strong label efficiency for this segmentation task.

The class-wise metrics show a consistent pattern: landing-site IoU is lower than background IoU for every SSL method. This indicates that visually similar terrain, incomplete regions, and uncertain boundaries remain the central segmentation challenges. MAE maintains the best overall balance across mIoU, pixel accuracy, Dice score, and landing-site IoU.

The difference between the supervised baseline and MAE is small in overall mIoU but remains meaningful for a safety-oriented application. The fully supervised model has access to more than four times as many labelled images, which gives it more direct examples of the subtle boundary cues that distinguish a landing site from similar background terrain. Nevertheless, the SSL ranking demonstrates that pretraining substantially narrows this gap before any dense labels are used.

TABLE VI
FINAL TEST PERFORMANCE. BOLD VALUES DENOTE THE BEST SSL RESULT.

Model	Labelled Images	mIoU	Pixel Accuracy	Mean Dice	Landing-Site IoU
DeepLabV3–ResNet50	1,416	0.8220	0.9148	0.9011	0.7609
SegFormer-B0	1,416	0.8154	0.9115	0.8969	–
YOLO26n-Sem	1,416	0.8041	0.9071	0.8897	–
MAE	314	**0.8055**	**0.9058**	**0.8908**	**0.7398**
SimCLR	314	0.8018	0.9031	0.8885	0.7364
DINOv2	314	0.7999	0.9010	0.8874	0.7366
BYOL	314	0.7843	0.8942	0.8773	0.7120

The ordering of the SSL models is also stable across the reported metrics. MAE has the highest mIoU, pixel accuracy, mean Dice, and landing-site IoU among SSL methods. This agreement matters because it avoids a conclusion based on a single metric: MAE does not obtain its lead merely through better background classification, but also provides the strongest overlap with the more difficult landing-site class.

A. Label-Efficiency Discussion

The fully supervised DeepLabV3–ResNet50 baseline used 1,416 labelled training images, whereas every SSL model used only 314 labelled images during downstream fine-tuning. The SSL methods therefore used 22.18% of the labelled data available to the baseline. Table VII reports the resulting label-efficiency comparison.

TABLE VII
LABEL-EFFICIENCY COMPARISON.

Method	mIoU	Gap	Recovered
DeepLabV3	0.8220	–	100.00%
MAE	0.8055	0.0165	98.00%
SimCLR	0.8018	0.0202	97.54%
DINOv2	0.7999	0.0221	97.31%
BYOL	0.7843	0.0377	95.41%

Table VII quantifies the trade-off between annotation cost and segmentation accuracy. MAE preserves 98.00% of the fully supervised mIoU while using only 22.18% of the labelled images, making it the most label-efficient SSL approach in this study.

MAE is only 1.65 percentage points below the supervised baseline despite a 77.82% reduction in labelled training images. SimCLR and DINOv2 also recover more than 97% of the baseline mIoU. These results demonstrate that the main benefit of SSL in this project is not surpassing a fully supervised model, but retaining most of its segmentation quality when dense annotations are scarce.

MAE is likely well suited to this task because reconstructing masked patches encourages the encoder to model global scene structure as well as local texture. Landing areas frequently depend on broad shape, continuity, and surrounding context, rather than a single distinctive object texture. SimCLR and DINOv2 also transfer effectively, but MAE provides the most consistent balance between background and landing-site overlap in the final evaluation.

From an annotation-cost perspective, the result is especially useful. A conventional supervised workflow requires a person to draw a dense mask around each landing-site region for all 1,416 training images. In the SSL workflow, the same 1,416 images are still valuable during pretraining even without masks; only the 314-image fine-tuning partition requires dense annotation. This separation explains why SSL is attractive when aerial imagery is plentiful but expert annotation time is limited.

V. ERROR ANALYSIS AND LIMITATIONS

Per-image IoU was used to select the five worst test images for each model. MAE shared none of these five images with the Part-A baseline, whereas SimCLR shared two. This shows that some difficult images are common across approaches, while others cause representation-specific failure patterns.

Figure 7 provides qualitative evidence for the low-IoU cases. Some predictions miss portions of a genuine landing-site region, whereas others classify visually similar terrain as a landing site. The diversity of these errors explains why mIoU remains below the fully supervised baseline even for the strongest SSL model.

Figure 8 complements the qualitative examples by identifying common difficult images. SimCLR shares two worst cases with the Part-A baseline, while BYOL and DINOv2 each share one. MAE shares none of its five worst cases with the fully supervised baseline, which indicates that its most difficult images have a different error profile. Across all methods, low-IoU examples often contain small landing regions, fragmented boundaries, or terrain with a similar colour and texture to the target class.

The Part-B test split was used for monitoring, checkpoint selection, and final reporting. This assignment design makes the reported SSL results slightly optimistic compared with an experiment using a completely untouched test set. A future study should preserve a separate downstream validation partition.

Further limitations should also be considered. The images originate from one aerial dataset and the binary labels do not explicitly represent operational constraints such as terrain slope, obstacle height, weather, aircraft type, or proximity to hazards. A segmentation mask is therefore a visual candidate

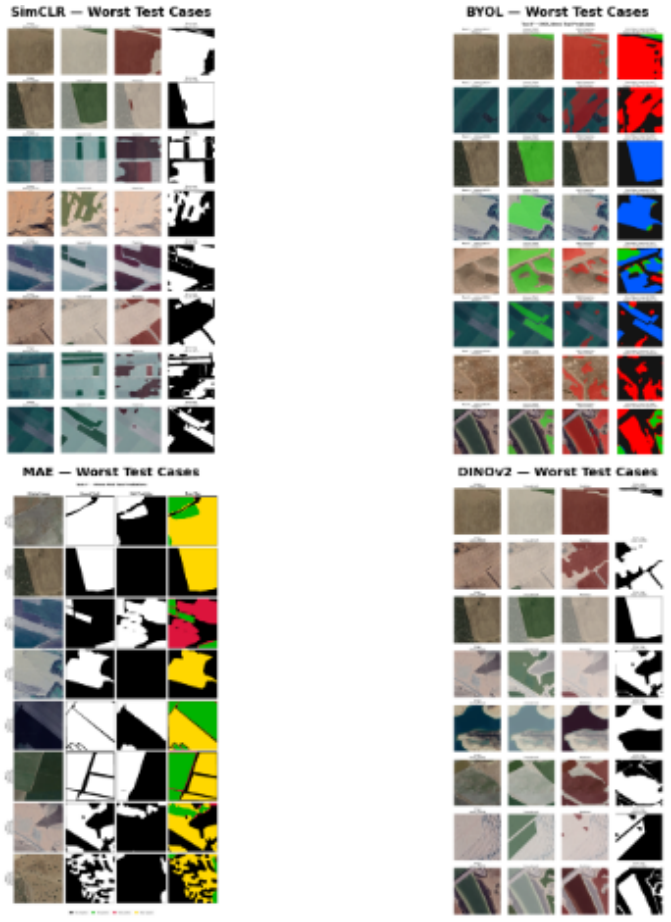


Fig. 7. Worst-performing test cases for the four SSL fine-tuned models. The examples show false-positive, false-negative, and boundary errors in the predicted landing-site masks.

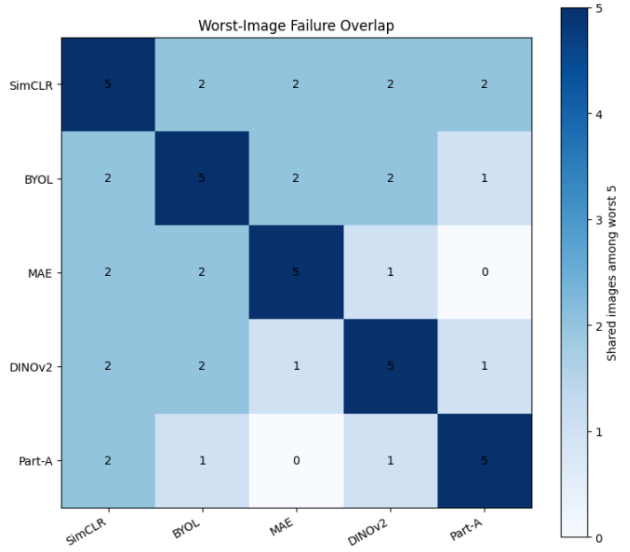


Fig. 8. Overlap among the five worst-performing test images. The diagonal value is five because each model is compared with its own worst cases.

map, not a final certification of landing safety. Future work could evaluate geographic transfer, use a fully held-out test set, and combine segmentation with additional geospatial safety information.

The error analysis should therefore be interpreted as a diagnostic rather than a complete safety assessment. False-positive regions would need further screening by a navigation or geographic-information system, while false negatives may be reduced by collecting more examples of small and irregular landing sites. Despite these limitations, the comparison identifies the types of imagery that should receive attention when improving the dataset or model design.

VI. Conclusion

DeepLabV3–ResNet50 achieved the strongest fully supervised result, with a test mIoU of 0.8220. MAE was the best SSL method, reaching 0.8055 mIoU using only 314 labelled fine-tuning images. It retained 98.00% of the supervised baseline mIoU while reducing labelled-data use by 77.82%. SimCLR and DINOv2 also retained more than 97% of the supervised mIoU, confirming that representation pretraining is valuable when dense labels are limited. The qualitative results show that fragmented boundaries and terrain resembling landing areas remain the principal errors. Overall, SSL pretraining is a practical strategy for reducing annotation cost while maintaining strong aerial semantic-segmentation performance.

References

[1] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," in *Proc. ICML*, pp. 1597–1607, 2020.

[2] J.-B. Grill et al., "Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[3] K. He et al., "Masked Autoencoders Are Scalable Vision Learners," in *Proc. CVPR*, pp. 16000–16009, 2022.

[4] M. Oquab et al., "DINOv2: Learning Robust Visual Features without Supervision," *arXiv:2304.07193*, 2023.